

Super-Li Inventory Management System

Operating Instructions

Version: adss2025_v01 | Module: Inventory

Course: Software Systems Analysis & Design — BGU

1. System Requirements

Before running the system, ensure the following are installed:

Requirement	Version	Notes
Java (JDK)	11 or higher	Required to run the JAR file
Operating System	Windows / Mac / Linux	Any OS with Java support
Terminal / Command Prompt	Any	To run the JAR

2. How to Run the System

Step 1 — Navigate to the release folder:

```
cd path/to/ADSS_Group_AV/release
```

Step 2 — Run the JAR file:

```
java -jar adss2025_v01.jar
```

Step 3 — The system will display the welcome screen:

```
=====
Welcome to Inventory Management
=====

1. Load sample data
2. Start empty
Choose:
```

Option 1 — Load sample data: Loads pre-configured categories, item types, items, and discounts. Recommended for testing all functionality.

Option 2 — Start empty: Starts the system with no data. You can add everything manually.

Note: There are no user accounts or passwords. The system starts immediately after choosing an option.

3. Main Menu Navigation

After startup, the main menu is displayed:

Option	Menu	Description
1	Category Operations	Add and view product categories
2	ItemType Operations	Manage product definitions (types)
3	Item Operations	Manage physical units of products
4	Discount Operations	Add and view promotions/discounts
5	Reports	Generate inventory and defective item reports
6	Alerts	View low stock alerts
0	Exit	Exit the system

4. Category Operations

Categories are organized in a tree structure with unlimited levels. A category can have any number of sub-categories. Items can only be assigned to leaf categories (categories with no children). Example: Dairy > Milk > 3% — where '3%' is the leaf.

4.1 Category Menu Options

Option	Action	Notes
1	Add root category	Creates a top-level category (e.g., Dairy)
2	Add sub-category	Creates a child category under an existing one
3	Show all categories	Lists all categories with full path
4	Show category by ID	Shows full details of a single category
5	Show root categories	Lists only top-level categories
6	Show category tree	Displays the full category hierarchy
0	Back	Return to main menu

Example — Adding a sub-category:

Select parent category:

1. Dairy

2. Dairy > Milk

Enter parent number: 1

Enter sub-category name: Milk

Sub-category 'Milk' added under 'Dairy'

5. ItemType Operations

An ItemType represents a product definition (e.g., 'Milk 3% Tnuva 1L'). It holds the product name, price, location, and minimum stock threshold.

5.1 ItemType Menu Options

Option	Action	Notes
1	Add item type	Define a new product with name, price, location
2	Show all item types	Lists all product definitions
3	Show item type by ID	Shows full details of one product
4	Update minimum quantity	Change the low-stock threshold
5	Show item types by category	Filter products by category
0	Back	Return to main menu

5.2 Required Fields When Adding an Item Type

Field	Description	Example
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Name	Product name	Milk 3% Tnuva 1L
Shelf number	Physical shelf number in store	1
Aisle number	Physical aisle number in store	1
Min quantity	Alert threshold for low stock	20
Cost price	Price paid to supplier	4
Selling price	Price charged to customer	6
Manufacturer	Name of manufacturer	Tnuva
Category	Leaf category from the tree	Dairy > Milk > 3%

6. Item Operations

An Item is a physical unit of a product. Each item tracks its location (shelf or warehouse), expiration date, and damaged status.

Option	Action	Notes
1	Add single item	Add one physical unit with expiration date
2	Add multiple items of same type	Add N units at once with same expiration
3	Show all items	Lists all physical units in the system
4	Show item by ID	Shows details of one unit
5	Show items by item type	Filter units by product type
6	Move one item to shelf	Move a warehouse item to shelf
7	Move multiple items to shelf	Move N items of a type to shelf
8	Move one item to warehouse	Move a shelf item to warehouse
9	Mark item as damaged	Flag an item as damaged
10	Unmark item as damaged	Remove damaged flag from an item
11	Update item expiration date	Change expiration date of an item
12	Remove item	Permanently delete a physical unit
13	Remove all defective items	Delete all damaged or expired units
0	Back	Return to main menu

Note: Dates must be entered in yyyy-mm-dd format (e.g., 2026-05-01). Expiration date is optional — leave blank and press Enter to skip.

Note: When removing items, if the total quantity of that product drops below the minimum threshold, a low stock alert will be displayed automatically.

7. Discount Operations

Two types of store promotions are supported: item-level discounts and category-level discounts. If multiple discounts apply to the same item, only the highest percentage is applied.

Option	Action	Notes
1	Add item discount	Apply % discount to specific products
2	Add category discount	Apply % discount to entire category
3	Show active discounts for item	View discounts active today for a product
4	Show all discounts	List all discounts in the system
5	Show final price for item type	Calculate price after best discount
0	Back	Return to main menu

8. Reports

Option	Report	Description
1	Category inventory report	Shows all items per selected category with quantities and prices
2	Show inventory by categories	Browse inventory filtered by one or more categories
3	Defective item report	Lists all damaged or expired items currently in the system
4	Purchasing report	Lists products below minimum threshold with recommended order quantities
0	Back	Return to main menu

9. Alerts

Alerts are generated automatically when a product's total quantity (shelf + warehouse) drops below its minimum threshold.

Option	Action	Notes
1	Show all alerts	Lists all products currently below minimum quantity
2	Show alert for item type	Check if a specific product has a low stock alert
0	Back	Return to main menu

10. Sample Data — Loaded on Startup

When selecting option 1 (Load sample data) at startup, the following data is pre-loaded into the system for testing all functionality.

10.1 Category Tree

```
Dairy (id:1)
  Milk (id:2) > 3% (id:3), 1% (id:4), 5% (id:5)
  Cheese (id:6) > Yellow (id:7), White (id:8)
  Butter (id:9) > 250g (id:10), 500g (id:11)
Hygiene (id:12)
  Shampoo (id:13) > 250ml (id:14), 500ml (id:15)
  Soap (id:16) > Bar (id:17), Liquid (id:18)
Beverages (id:19)
  Juice (id:20) > Orange (id:21), Apple (id:22)
  Water (id:23) > 500ml (id:24), 1.5L (id:25)
Bakery (id:26)
  Bread (id:27) > White (id:28), Whole Wheat (id:29)
Snacks (id:30)
  Chips (id:31) > Regular (id:32), Baked (id:33)
  Chocolate (id:34) > Dark (id:35), Milk Choc (id:36)
```

10.2 Item Types (Products)

ID	Name	Category	Cost	Price	Min Qty	Manufacturer
1	Milk 3% Tnuva 1L	Dairy>Milk>3%	4	6	20	Tnuva
2	Milk 1% Tnuva 1L	Dairy>Milk>1%	4	6	20	Tnuva
3	Milk 5% Tnuva 1L	Dairy>Milk>5%	4	7	15	Tnuva
4	Yellow Cheese 200g	Dairy>Cheese>Yellow	8	12	10	Tnuva
5	White Cheese 200g	Dairy>Cheese>White	6	9	10	Tnuva
6	Butter 250g	Dairy>Butter>250g	6	9	8	Tnuva
7	Butter 500g	Dairy>Butter>500g	11	16	5	Tnuva
8	Pinuk Shampoo 250ml	Hygiene>Shampoo>250ml	8	14	8	Pinuk
9	Pinuk Shampoo 500ml	Hygiene>Shampoo>500ml	14	22	5	Pinuk
10	Dove Soap Bar	Hygiene>Soap>Bar	3	6	10	Dove
11	Dove Soap Liquid	Hygiene>Soap>Liquid	7	12	8	Dove
12	Prigat Orange Juice	Beverages>Juice>Orange	4	7	15	Prigat
13	Prigat Apple Juice	Beverages>Juice>Apple	4	7	15	Prigat
14	Water 500ml	Beverages>Water>500ml	1	3	30	Neviot
15	Water 1.5L	Beverages>Water>1.5L	2	5	20	Neviot
16	White Bread	Bakery>Bread>White	3	6	10	Berman
17	Whole Wheat Bread	Bakery>Bread>Whole Wheat	4	7	10	Berman
18	Regular Chips 100g	Snacks>Chips>Regular	3	6	20	Elite

19	Baked Chips 100g	Snacks>Chips>Baked	4	7	15	Elite
20	Dark Chocolate 100g	Snacks>Choc>Dark	5	9	10	Elite
21	Milk Choc 100g	Snacks>Choc>Milk	4	8	12	Elite

10.3 Physical Items (Units)

Item Type	Qty on Shelf	Qty in Warehouse	Expiration Date	Notes
Milk 3% Tnuva 1L (id:1)	2	2	2026-05-01 / 2026-05-15	1 unit is damaged
Milk 1% Tnuva 1L (id:2)	1	1	2026-05-01	
Yellow Cheese 200g (id:4)	1	1	2026-06-01	
Pinuk Shampoo 250ml (id:8)	1	1	2027-01-01	
Water 500ml (id:14)	2	1	2026-12-01	
Whole Wheat Bread (id:17)	2	0	2026-04-25	

10.4 Active Promotions

Type	Target	Discount	Start Date	End Date
Category Discount	Dairy (id:1) — all sub-items	15%	2026-04-01	2026-04-30
Category Discount	Hygiene (id:12) — all sub-items	10%	2026-04-15	2026-04-25
Item Discount	Milk 3%, Milk 1%, Milk 5% (ids:1,2,3)	20%	2026-04-18	2026-04-20